Operations Research project
ING1 - INT 3 – 2025/2026
Groupe 3

Complexity Report

Somphone Isabelle, Monsallier Corentin, Duret Maxime, Bou Ferraa Karel, Ouazari Judia

Mathematics Department Efrei Paris **BENMERZOUK Djamila**

Introduction

To complete this semester, we were asked to realize a project for the operation research class that put into action the things that we learned and make us work on specific scenario of transportation problem. Our group chose to realize the project with python as a programming language and use GitHub for collaboration.

Our goal was to build a program that finds the cheapest way to transport goods from suppliers to customers. We implemented the North-West Corner and Balas-Hammer methods to start the resolution, and the Stepping-Stone algorithm to find the final optimum. To handle the graph logic, we used Breadth-First Search (BFS) to detect cycles and ensure the results were valid.

More than just coding the solution, we also focused on complexity analysis. We tested our algorithms with random problems of different sizes, from 10 up to 1000 iterations, to see how they perform in real-world conditions. More iterations were too long to compute with our code which made us stop at 1000.

Our situation

We have four main algorithms to run:

- **Balas hammer :**

The Balas-Hammer solution allows to generate an initial feasible solution. At each iteration, it computes penalties for each row and column that corresponds to the difference between the two smallest costs. The algorithm selects either the row/column with the highest penalty and assigns as much quantity as possible to its lowest_cost cell.

- **North west :**

This method starts at the top-left corner of the cost matrix and allocates as much quantity as possible based on the remaining supply and demand. The process will move either down or to the right depending on the situation. It does not take into account transportations costs.

- **Stepping stone with balas hammer:**

When the initial solution is generated using the Balas-Hammer method, the stepping-stone method is applied in the same way to reach optimality. However, because Balas-Hammer already provides a better starting solution, fewer iterations are generally needed.

- **Stepping stone with north west**

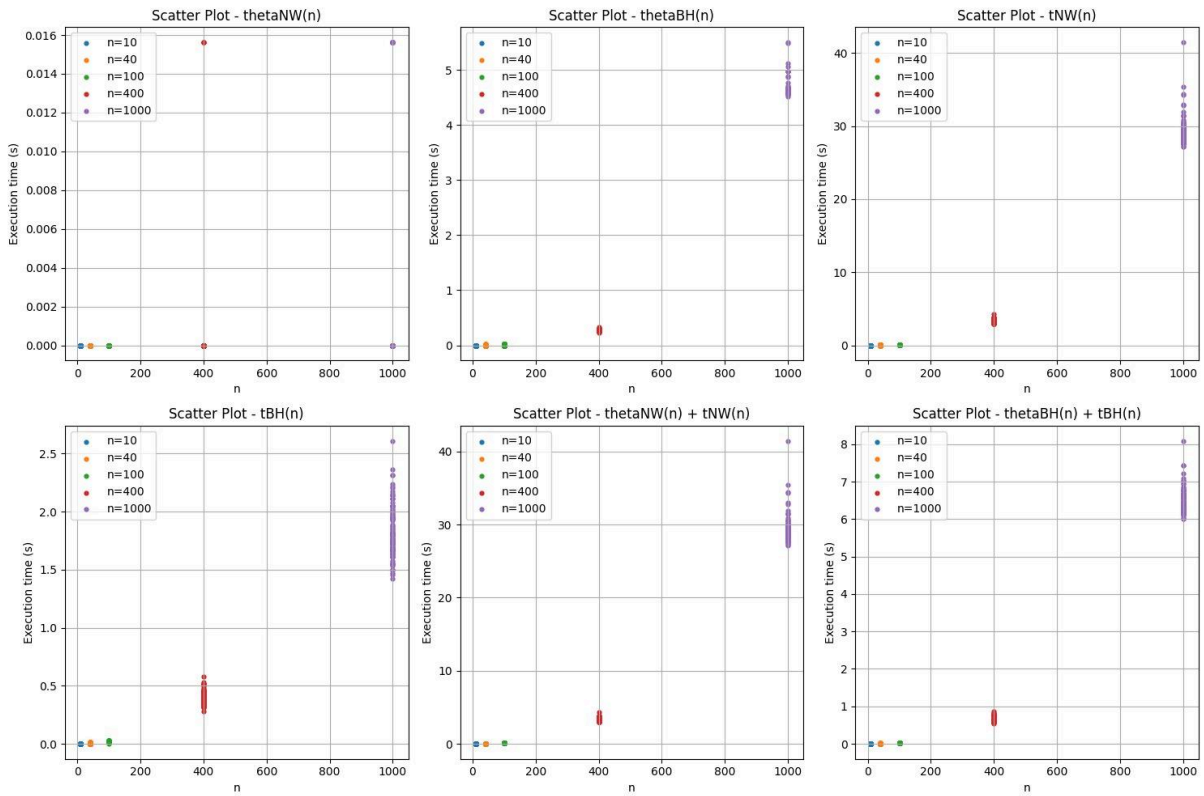
The algorithm evaluates unused cells by constructing closed cycles and computing marginal costs. If a negative marginal cost is found, the solution is adjusted along the corresponding cycle to reduce the total cost.

Doing so, we can estimate that the different measures are at first :

- $\theta_{NW}(n)$: execution time for North-West algorithm
- $\theta_{BH}(n)$: execution time for Balas-Hammer algorithm
- $t_{NW}(n)$: execution time for Stepping-Stone with North-West initial solution
- $t_{BH}(n)$: execution time for Stepping-Stone with Balas-Hammer initial solution

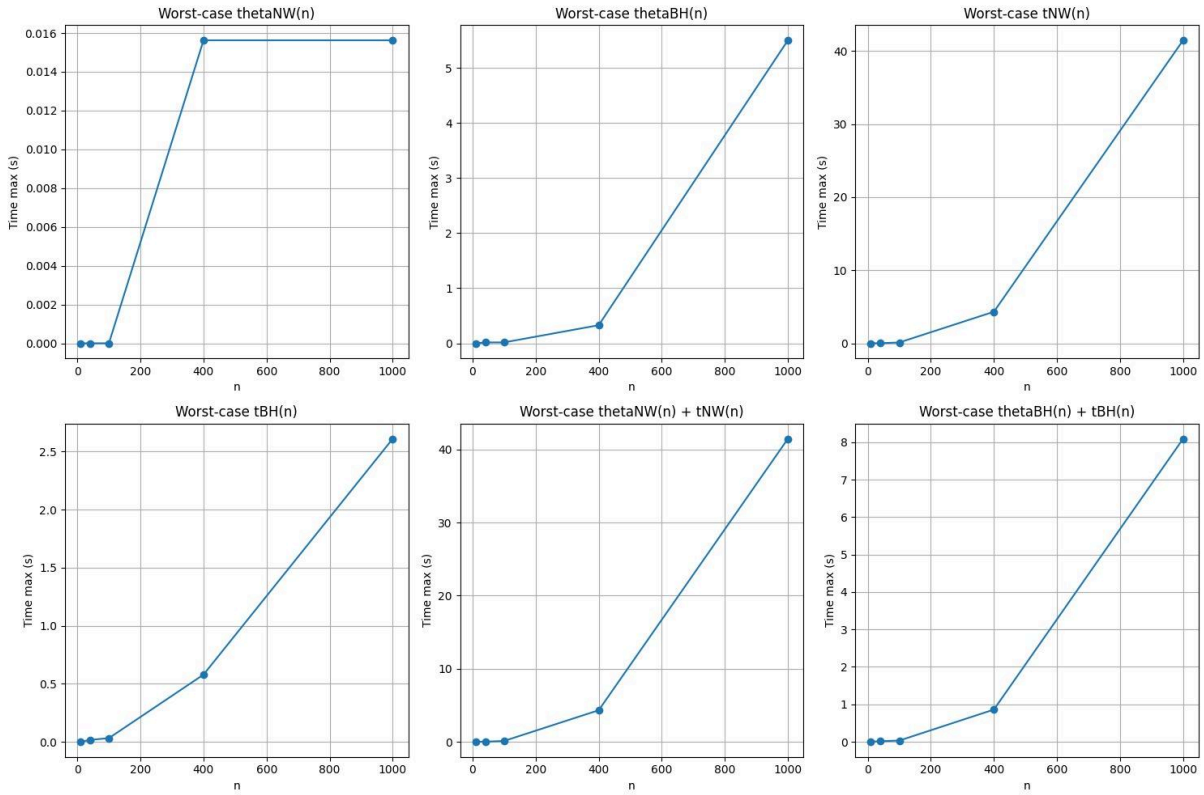
First to visualize the different methods and algorithms we realize some scatter plot. Those plots show that the execution time increases significantly when the transport square problem size n becomes larger, especially from n=400 to n=1000. For small values, the execution times are very close to zero for all algorithms, which means the computational cost is nearly totally negligible at this scale. Therefore, when n increases, the difference between the different methods becomes much more visible. The North-West algorithm θNW(n) remains extremely fast even for large sizes, while the Balas-Hammer algorithm θBH(n) becomes much slower. This suggests that North-West has a much lower initialization cost than Balas-Hammer.

The same behavior appears for the Stepping-Stone method that is used with both methods. When the initial solution comes from North-West tNW(n), the total execution time becomes the highest of all plots, especially for n=1000, where it reaches more than 40 seconds. In comparison, using Balas-Hammer as the initial solution tBH(n) produces a much lower execution time. This means that even if Balas-Hammer is slower at the beginning, it provides a better starting solution, which reduces the optimization time of Stepping-Stone. For the worst-case analysis, we will therefore focus on the upper envelope of these scatter plots by taking the maximum execution time for each nnn. This will help us estimate the practical worst-case complexity and compare which method is more efficient for large-scale problems.



Worst-case scenario

Following is the plot of each longest run for each requested algorithm :



We consider the worst-case as the maximum execution time observed for each problem size. The NW method remains very fast, while BH is slower due to penalty computations. However, when combined with the Stepping-Stone method, BH leads to fewer iterations and better overall performance. In contrast, NW often results in higher total execution time due to slower convergence.

Complexities :

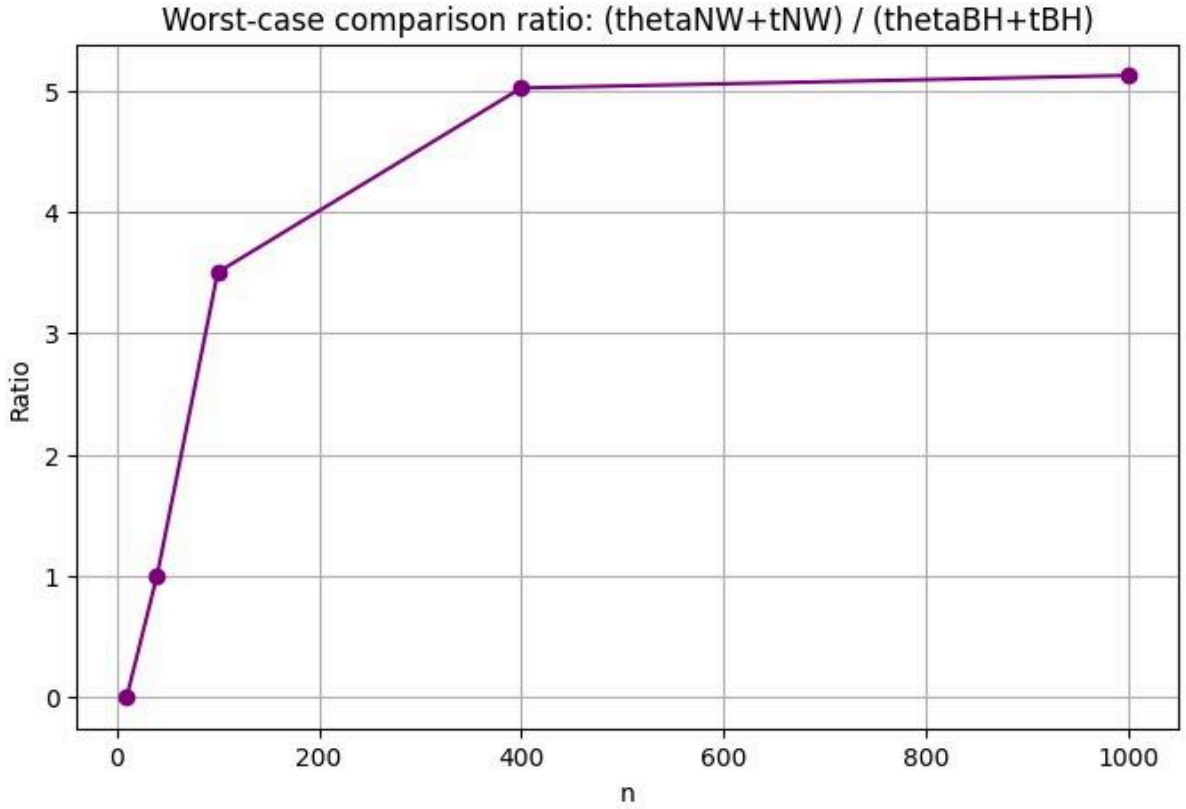
- $\theta_{NW}(n)$: $O(n)$
- $\theta_{BH}(n)$: $O(n^2)$
- $t_{NW}(n)$: $O(n^3)$
- $t_{BH}(n)$: $O(n^2)$
- $\theta_{NW}+t_{NW}(n)$: $O(n^3)$
- $\theta_{BH}+t_{BH}(n)$: $O(n^2)$

Ratio

The graph illustrates the worst-case performance ratio. We can see that when the ratio is more than 1, the North-West method is slower than the Balas-Hammer method. In the graph, as soon as n exceeds 40, the ratio climbs rapidly above 1, reaching approximately 5 when n = 1000.

Even though the Balas-Hammer algorithm is more complex to calculate initially ($\theta_{BH} > \theta_{NW}$), it provides a starting solution that is much closer to the optimal result.

The more values our use case has, the more efficient Balas-Hammer method will become. The ratio's complexity will be O(n).



Conclusion

In conclusion, the worst-case analysis suggests that the North-West algorithm behaves approximately like O(n2) when compared to the graph, since its execution time increases regularly and remains relatively low even for large values of nnn. In contrast, the Balas-Hammer algorithm seems closer to O(n3), as its execution time grows much faster when n increases. This shows that North-West is more efficient for quickly generating an initial feasible solution, while Balas-Hammer is more expensive in terms of computation time but can provide a better starting point for optimization methods such as Stepping-Stone.